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OVERVIEW

- Characteristics of Early Childhood
- Physical development
- Cognitive development
- Socio-emotional development
- Moral development

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EARLY CHILDHOOD – MAIN CHARACTERISTICS

- Age 2-5 years
- From end of infancy to beginning of school years
- Toddler and pre-school years
- Some key developmental tasks:
 - Improved gross and fine motor coordination
 - Improved executive functioning (pre-frontal cortex)
- Sustained attention
- Language development
- Develop pre-operational thought (Piaget) – represent the world with words, images and drawings
- Increased ability to consider others' perspectives

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GROWTH AND PHYSICAL CHANGES

- Growth slower than it was in infancy – yet still significant
- Height increases 6-7 cm per year
- Weight increases by about 2kg per year
- Girls only slightly smaller and lighter than boys
- Girls slightly more fatty tissue; boys slightly more muscle

Table 2.1 Average heights and weights for girls and boys during early childhood (3-8 years old)

Age	Girls' height (cm)	Boys' height (cm)	Girls' weight (kg)	Boys' weight (kg)
3	94	95	14	14.5
4	101	102.5	16	16.5
5	107.5	109	18	18.5
6	114.5	115.5	20	21
7	121.5	122	23	23.5
8	127.5	128	25.5	26

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GROWTH AND PHYSICAL CHANGES

- Growth patterns vary from one child to the next, according to genetic inheritance and environments
- Parents' genes
- Ethnicity
- Socioeconomic status
- Nutrition

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GROWTH AND PHYSICAL CHANGES

- Body lengthens and slims down; less body fat by end of early childhood
- Portion of head to body is less than in infancy (less "top-heavy")
- 20 teeth by about 3 years old
- Bowel and bladder control

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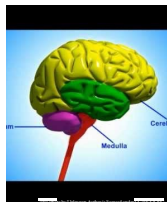


BRAIN AND NERVOUS SYSTEM DEVELOPMENT

- Brain becomes further organized to support mental and physical behaviours in which child engages
- Increased number and size of dendrites
- Dramatic increase in volume for some parts of brain – especially frontal cortex.
- Volume less in other parts due to synaptic pruning – so overall size of brain does not change much
- Myelination of axons continues, increasing speed and efficiency of transmissions

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BRAIN AND NERVOUS SYSTEM DEVELOPMENT



- Changes in brain and nervous system lead to advances in functioning of pre-frontal cortex and other parts of the brain
- Changes enable children to better
 - Plan their actions
 - Attend to stimulus
 - Develop language

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GROSS MOTOR SKILLS

- Increased skills involving large-muscle activity
- Move around with greater confidence; less effort
- Enjoy hopping, jumping, climbing, and running back and forth
- May master monkey bars
- May learn to ride a bike or swim

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FINE MOTOR SKILLS

- Fine-tuned movements of small muscles in fingers/hand improves - clumsiness decreases and precision increases
- Able to pick up tiny objects between thumb & forefinger
 - Stack blocks & work on jigsaw puzzles
 - Turn pages of book
 - Use crayons and pencils for drawing and writing
 - Self-feed (with spoon and fork) & get dressed (zip up zips)

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FINE MOTOR SKILLS: EXAMPLE ART



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GROSS AND FINE MOTOR SKILLS

Approximate age	Gross motor skill	Fine motor skill
2.5-3.5 years	Walks competently; runs in straight line; jumps in air with both feet	Copies a circle; scribbles; can use eating utensils; stacks a few small blocks
3.5-4.5 years	Walking stride 80 per cent of adult; runs at one-third adult speed; throws and catches large ball, but stiff-armed	Copes with large buttons; copies simple shapes; makes simple representational drawings
4.5-5.5 years	Balances on one foot; runs reasonable distances without falling; can swim for short distances	Uses scissors; draws people; copies simple letters and numbers; builds complex structures with blocks

Note: The ages given above are approximate, and skills vary with the life experiences available to individual children and with the situations in which the skills are displayed.
Source: Adapted from Kalverboet, Hopkins, and Geuze (1993).

- Play contributes to development of gross and fine motor skills

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HEALTH & WELLNESS: ILLNESS & ACCIDENTS

- Generally a healthy period of life
- Accidents are leading cause of death in NZ (e.g. drowning, falls)
- Preventable infectious diseases leading cause of death in developing world
- Poverty negatively impacts health (malnutrition, inadequate access to health care, unsafe water)
- Nutrition/eating habits, and exercise, affects skeletal growth, body shape, & disease susceptibility
 - Eating habits shaped by caregiver behaviour
 - Young children need 15 mins physical activity per hour

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COGNITIVE DEVELOPMENT



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COGNITIVE DEVELOPMENT

- During early childhood, children are creative, have active imaginations, and their mental representation of the world is improving
- Changes in the brain enable children to plan their actions, attend to stimuli more effectively, and to make strides in language development
- Contextual factors important too, e.g., poverty, parenting quality, maternal sensitivity

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PIAGET: PRE-OPERATIONAL STAGE

Piaget's Periods of Cognitive Development			
Birth to 2 years	Sensori-motor	Uses senses and motor skills, items known by use	Object permanence learned
2-6 yrs	Pre-operational	Symbolic thinking, language used; egocentric thinking	Imagination/experience grow, child de-centers
7-11 yrs	Concrete operational	Logic applied, has objective/rational interpretations	Conservation, numbers, ideas, classifications
12 yrs to adulthood	Formal operational	Thinks abstractly, hypothetical ideas (broader issues)	Ethics, politics, social/moral issues explored

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PIAGET: PRE-OPERATIONAL STAGE

Pre-Operational Stage (Symbolic Function & Intuitive Thought Sub-Stages)

- Lasts from 2 to 7 years
- Pre-school children begin to represent the world with words, images, drawings, and pretend play
- Pre-school children are egocentric – unable to see another person's perspective well; assume others see, hear, feel, and share same point of view as their own

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PIAGET: PRE-OPERATIONAL STAGE

- Video about the mountain task – which demonstrates egocentrism



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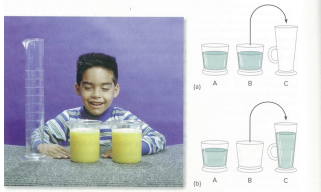
PIAGET: PRE-OPERATIONAL STAGE

- Pre-school children are animistic – they believe that all things are living (e.g., toys)
- Hold magical beliefs
- Begin to use early reasoning and want to know the answers to all sorts of questions (“why?”)
- Only able to focus attention on one characteristic of an object, to the exclusion of other characteristics (centration)

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PIAGET: PRE-OPERATIONAL STAGE

- Centration affects ability to understand law of conservation



- Also due to inability to mentally reverse actions

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PIAGET: PRE-OPERATIONAL STAGE

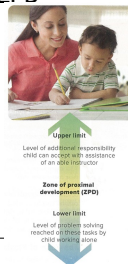
- Video about the conservation during early childhood



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VYGOTSKY: SOCIAL INTERACTION & ZPD

- Vygotsky emphasises **social contexts** of learning
- Cognitive development and learning occurs through **interactions** with others e.g., family, teachers, and peers
- **Zone of proximal development** – level of competence that is achievable with assistance/guidance from capable others
- Guided support useful (scaffolding)
- Private speech /self-talk considered useful for problem solving – later becomes inner speech



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INFORMATION PROCESSING APPROACH

- Children's ability to pay **attention** improves during pre-school years.
 - Better able to focus mental resources on select information
 - Better able to allocate attention to a goal
 - Better able to **sustain attention** on a task or event
- However, they attend to what stands out even if it is not relevant to solving a problem

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INFORMATION PROCESSING APPROACH

- Short-term memory improves
 - Can remember more items presented to them 30 seconds later
- Long-term memories become more accurate
 - However, susceptible to suggestion
- Autobiographical memory develops
 - Remember significant events and personal experiences

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INFORMATION PROCESSING APPROACH

- Executive function improves due to changes to prefrontal cortex.
- Executive function involves managing one's thoughts to engage in goal-directed behaviour and to exercise control. Advances in:
 - cognitive inhibition (e.g., inhibiting an incorrect tendency)
 - cognitive flexibility (shifting attention to another item)
 - goal-setting (mastering a skill)
 - delaying gratification (ability to forgo an immediate reward for a more desirable one later)
- Important for school readiness
- Also influenced by adult role-models; having good self-regulators around them

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INFORMATION PROCESSING APPROACH

- Video about executive function



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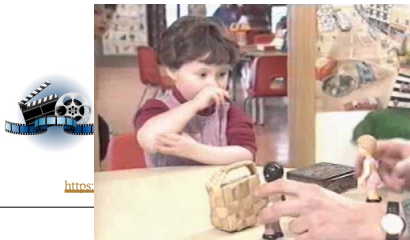
THEORY OF MIND

- Refers to awareness of mental processes of self and others
- Over the pre-school years, children:
 - Better distinguish between different emotions
 - Realise that what others see in front of them is not necessarily what they see
 - Learn that events can be inaccurately represented – that people can have false beliefs
- Theory of mind is tested in the 'Sally and Anne task'

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INFORMATION PROCESSING APPROACH

- Video about theory of mind



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THEORY OF MIND

- What contributes to development of theory of mind?
 - Advances in prefrontal cortex functioning
 - Advances in language skills
 - Social interactions
 - Having parents, siblings, and friends who talk about mental states and emotions (e.g., "Are you feeling angry that ..."
 - "What do you think he is thinking?")
- Engaging in pretend play with peers

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LANGUAGE DEVELOPMENT

- Length of sentences increases
- Vocabulary dramatically increases - one word per hour; 14,000 words by the end of early childhood (semantics)
- Increasingly capable of producing sounds of language (phonology)
- Learn how words should be ordered (syntax)
- Learns the rules of language (morphology), for example:
 - Use prepositions (on, in)
 - Use plural form of nouns (e.g., pencils)
 - Use possessives (e.g., cat's, brother's)
 - Use past tense (walked)
 - Overgeneralise rules (e.g., foots, tooths; goed instead of went)

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SOCIAL-EMOTIONAL DEVELOPMENT



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DEVELOPMENT OF SELF-UNDERSTANDING

- Children in early childhood are developing a sense of themselves
- Young children describe themselves, and compare themselves to others, in terms of physical characteristics, material characteristics, and activities (but initially not psychological characteristics)
 - *"I have brown hair and Lucia has blonde hair", "I have a bicycle", "I am taller than Lucia"*

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DEVELOPMENT OF SELF-UNDERSTANDING

- At age 4 to 5, psychological traits and emotions are also used in self-descriptions
 - *"I am always happy"*
- Self-descriptions are unrealistically positive – because:
 - Cannot yet distinguish between actual competence and desired competence
 - Do not yet compare their abilities with peers
 - Compare present abilities with abilities when younger

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UNDERSTANDING OTHERS

- As theory of mind develops, pre-school children come to understand that others have emotions and desires
- At age 4-5 years, children start describing others in terms of psychological traits too
 - *"Lucia is nice/kind"*
- Pre-school children not as egocentric as previously thought; somewhat socially aware and sensitive

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EMOTIONAL DEVELOPMENT

- Use more terms to describe emotions
- Experience self-conscious emotions like grief and guilt
 - Influenced by parents responses to children's behaviour, e.g., *"You should feel bad that you snatched that toy off ..."*
- Better understand causes & consequences of emotions
- Better understand that the same event can elicit different emotions in different people
- Increasingly aware that they need to manage their emotions to meet social expectations

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INITIATIVE VS GUILT (ERIKSON)

- Initiative vs guilt is the third stage of eight in Erikson's stages of psychosocial development (occurs 3-5 years)



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INITIATIVE VS GUILT (ERIKSON)

- At ages 3 to 5, children seek 'purpose' and initiate ideas and experiences to please themselves and others
- If others respond with enthusiasm, this leads to a sense of purpose, and the child continues to show initiative
- If interactions are harsh (i.e., when child oversteps social boundaries), the child may feel guilt and remorse, and refrain from showing initiative and leadership in future.



Guilt

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EMOTIONAL DEVELOPMENT

- Home-life affects expression and regulation of emotions
 - Children use more positive emotion words when there is a high incidence of positive emotion, and a low incidence of negative emotion, at home
 - Children better at regulating negative emotion and self-soothing when parents take an emotion-coaching approach (versus an emotion-dismissive approach)

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EMOTIONAL DEVELOPMENT

- Ability to regulate emotions benefits child's social competence and peer relationships
 - Moody and emotionally negative children experience more rejection
 - Emotionally positive children are more popular
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SOCIAL DEVELOPMENT

- Good peer relationships promote social-emotional development
 - Frequency of peer interactions (both positive and negative) increases
 - By age 3, children already show a preference for spending time with same-sex playmates – and this preference increases between 4 to 6 years (& beyond)
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MORAL DEVELOPMENT (PIAGET & KOHLBERG)

- From ages 4-7 years, children:
 - Believe rules are created by wise adults, must be followed, and cannot be changed
 - Believe breaking a rule leads to punishment
 - Focus on consequences of behaviour, not the intention behind behaviour
 - Pre-school children obey because adults tell them to obey. Moral decisions based on fear of punishment
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MORAL DEVELOPMENT

- A **behavioural and social cognitive approach** argues that children's behaviour is influenced by:
 - Role models around them
 - Reward and punishment
- Ability to resist temptation is closely related to self-control and ability to delay gratification
- What children do in one situation is only weakly related to what they will do in another situation
- A totally honest child is virtually non-existent

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MORAL DEVELOPMENT: SECRET LIFE OF 4, 5, & 6 YEAR OLDS

- Documentary clip - learning to lie



- <https://www.youtube.com/watch?v=...>



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EARLY CHILDHOOD ECOLOGY



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PLAY

- Preschool years are the “golden age” of **imaginative/pretense play** (peaks at 4-5 years)
- Social play (with other children) increases



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PLAY

- Play promotes physical development
 - Gross and fine motor skills
- Play promotes cognitive development:
 - Abstract thinking, imagination, attention, problem-solving, perspective-taking, language
- Play promotes social-emotional development:
 - Enjoyment, relaxation, self-expression, cooperation, sharing, turn-taking, self-confidence

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ATTACHMENT

- Attachment is the close emotional bond between two people; warmth, responsiveness, attunement, synchrony.
- Attachment between infant/child & parent/ caregiver has a survival function & promotes:
 - Sense of security
 - Regulation of affect & arousal (strategies for self-soothing internalised)
 - Expression of feelings & communication
 - Secure base to explore

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ATTACHMENT

- Child develops an internal working model of attachment
 - A simple mental model of the caregiver, their relationship with him or her, and the self as deserving on nurturing and care
- Attachment influences child's responses to other people, emotional understanding and self-concept
 - Higher levels of maternal sensitivity in early childhood associated with higher total brain volume
 - Secure attachment linked to better executive function

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PARENTING QUALITY

- Quality parenting important for children's development
 - Sensitivity, warmth, & support important
 - Authoritative parenting style most beneficial to child

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PARENTING STYLES



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PARENTING QUALITY

- Abuse, maltreatment, neglect detrimental for development
 - Higher risk of poor emotion regulation, attachment problems, problems in peer relationships, difficulty adapting to school, depression
 - Important to discover how to prevent child maltreatment and to intervene in children's lives when they have been maltreated
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SUMMARY

- Some important features of early childhood features:
 - Improved gross and fine motor skills
 - Improved executive function linked to growth of pre-frontal cortex
 - Improvements in sustained attention
 - Language development
 - Develop pre-operational thought (Piaget) – represent the world with words, images, drawings
 - Erikson's initiative vs guilt stage
 - Increased ability to consider others' perspectives
 - Parenting quality, secure attachment, and play promote aspects of development
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